Why doesn't the model diverge?

The Gaussian null regime and the case for structural divergence

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A diagnosis of *Changing the Networked Mind* (v1 $\rightarrow$ v3)

The puzzle: where is the divergence?

A model of science should be able to show:

- **schools** — communities that disagree and *stay* disagreeing
- **schisms** — a field splitting under contact, not despite isolation
- **incommensurability** — camps that cannot hear each other
- a **lone discoverer** whose new commitment survives the crowd

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Every configuration, every topology, every parameter setting
contracts to consensus. No schools. No schism. Ever.

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This talk: that is not a tuning problem. It is a theorem about the representation.

The model in one slide

Each agent i holds a Gaussian belief net in **information form**:

$$q_i(\boldsymbol{\varphi}) \;\propto\; \exp\!\Big(-\tfrac{1}{2}\,\boldsymbol{\varphi}^\top \boldsymbol{\Pi}_i\,\boldsymbol{\varphi} + \boldsymbol{h}_i^\top \boldsymbol{\varphi}\Big)$$

one precision matrix $\boldsymbol{\Pi}_i$, one potential $\boldsymbol{h}_i = \boldsymbol{\Pi}_i \boldsymbol{\mu}_i$

Learning (Fisher accumulation)

$$\boldsymbol{\Pi}_i \leftarrow \boldsymbol{\Pi}_i + \boldsymbol{J}, \qquad \boldsymbol{h}_i \leftarrow \boldsymbol{h}_i + \boldsymbol{j}$$

every observation deposits precision; the mean is $\boldsymbol{\mu}_i = \boldsymbol{\Pi}_i^{-1} \boldsymbol{h}_i$

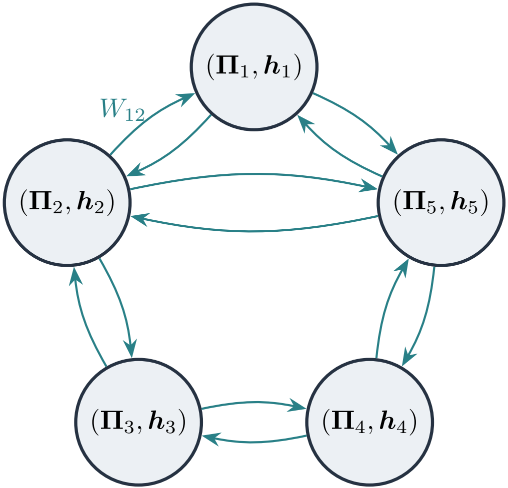
Communication (trust-weighted fusion)

$$\boldsymbol{\Pi}_i \leftarrow \sum_j W_{ij}\,\boldsymbol{\Pi}_j, \qquad \boldsymbol{h}_i \leftarrow \sum_j W_{ij}\,\boldsymbol{h}_j$$

W row-stochastic on the trust graph; precision-weighted averaging of the *whole* belief object

Off-diagonal Π_{uv} = represented coupling between commitments (the "wiring"); structural moves (expansion, Bayesian model reduction) edit which entries exist.

Every exchange is averaging of everything



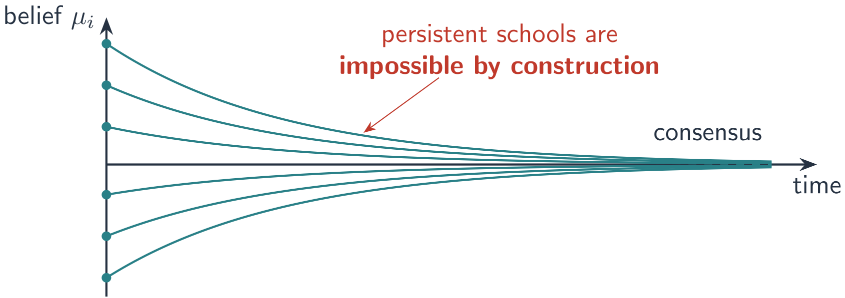
Agents exchange their **entire** $(\mathbf{\Pi}, \boldsymbol{h})$ and replace their own with a precision-weighted average.

No agent stores *whose* precision it absorbed — sources are anonymous once fused.

Theorem 1: pooling is a contraction

Pooling is a contraction[1]

On a connected trust graph, precision-weighted averaging contracts inter-agent disagreement **geometrically** toward consensus — *independent of content*.



Distrust (W_{ij} small) only rescales the exponent. It buys *time*, never a wall.

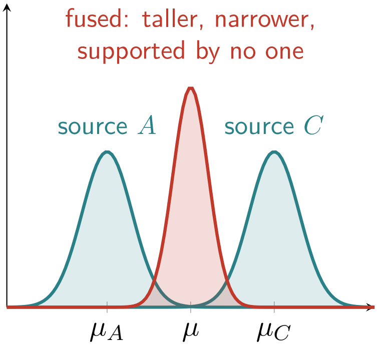
[1] DeGroot (1974); contraction under fixed row-stochastic trust formalised in Catal (2024).

Theorem 2: Gaussian conflict resolves by compromise

Fuse two *contradictory* sources:

$$\mathbf{\Pi} = \mathbf{\Pi}_A + \mathbf{\Pi}_C, \quad \boldsymbol{\mu} = \mathbf{\Pi}^{-1}(\mathbf{\Pi}_A \boldsymbol{\mu}_A + \mathbf{\Pi}_C \boldsymbol{\mu}_C)$$

- mean lands **between** the sources
- precision **adds** — confidence *grows* while averaging contradictions



[2] On conflict-blind precision pooling see O'Hagan (2012), *combining probability distributions*.

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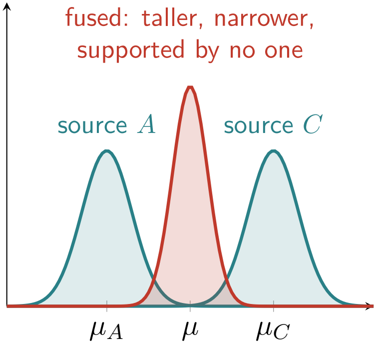
- mean lands **between** the sources
- precision **adds** — confidence *grows* while averaging contradictions
- the result is a confident belief in a configuration neither source supports

Light tails cannot reject extreme sources; conflict is invisible to the update.[2]



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The missing slot

There is **no coordinate in** $(\boldsymbol{\Pi}, \boldsymbol{h})$ whose value can mean "torn between two hypotheses."

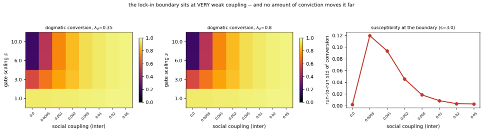
What a model of science needs	**What the Gaussian form can store**
rival hypotheses, held simultaneously	one mean $\boldsymbol{\mu}$
represented disagreement, a "hung jury"	one precision $\boldsymbol{\Pi}$ (always unimodal)
suspended judgment under conflict	conflict $\Rightarrow$ more confidence
knowing *who* disagrees, *about what*	sources erased at fusion

This is impoverishment **by representation**, not by parameters: the mean-field/Gaussian choice pulls every tension to the mean *before* dynamics can act on it.

What this does to a model of science

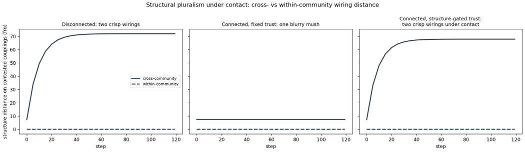
Corollaries of the two theorems:

- **The lone discoverer is averaged away.** A commitment held by one agent is diluted $1/N$ per fusion round — discovery cannot survive its own community.
- **No denial, no crisis.** Anomalies just shift the mean smoothly; there is no regime where evidence is resisted and then breaks through.
- **Contact is always fatal to the minority.** Lock-in survives only by *isolation*; any nonzero coupling converts the smaller camp.

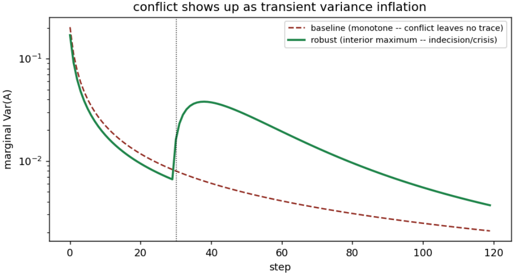


Lock-in boundary: the dogmatic camp persists only in the isolated corner; faint contact collapses it.

The simulations do exactly what the theorems say



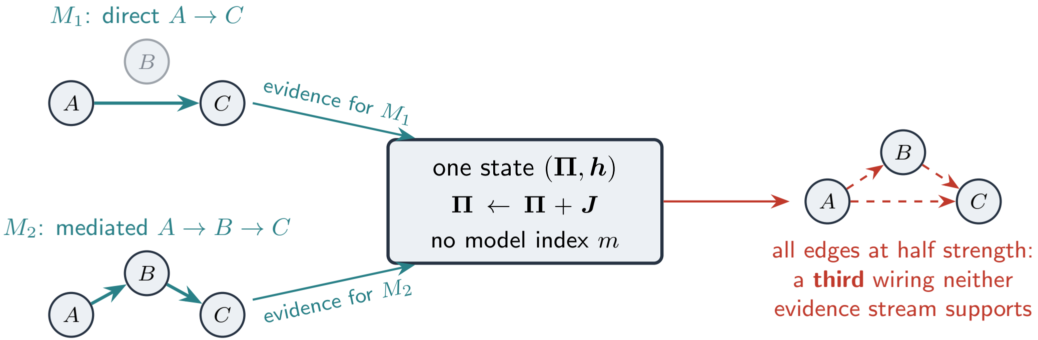
Inter-camp distance $D(t)$, log scale: under fixed trust (baseline), divergence decays **exponentially** — the contraction, measured.



Conflicting sources: baseline fusion averages them with opposite sign and *gains* confidence — Theorem 2, measured.

No parameter setting escapes this. The null result is the representation speaking.

The capability that is missing: holding two nets at once



The state is $q(\boldsymbol{\varphi})$, not $q(\boldsymbol{\varphi}, m)$. The evidence that *discriminates* the wirings — $I(o; m) = \mathrm{JSD}\big(p(o \mid M_1),\, p(o \mid M_2)\big)$ per observation — has nowhere to accumulate; it is erased at deposit time. The social failure (fusing two neighbours' nets) is the same collapse, between heads instead of within one.

This is not just our model

The same wall appears wherever beliefs are pooled under a Gaussian / mean-field form:

- **Federated inference**[3] — *prescribes* precision-weighted pooling, justified by a common-cause assumption. Where that assumption fails (rival paradigms), pooling is exactly the wrong move — and the formalism cannot express its failure.
- **Opinion dynamics** — DeGroot-style averaging contracts under any fixed trust; centuries-of-disagreement requires breaking the averaging, not reweighting it.[4]
- **Structure learning** — Bayesian model comparison[5] scores rival structures, but each agent still *owns a single posterior* — rivals live in the analyst's ledger, not in any agent's head.

The shared root: the mean approximation has no slot for a represented rival.

[3] Friston et al. (2023), *Federated inference and belief sharing*.

[4] DeGroot (1974); Catal (2024).

[5] Smith et al. (2022), structure learning as Bayesian model comparison; Friston et al. (2018), Bayesian model reduction.

What would be required

Structured learning over **hypothesis spaces**, not over one precision matrix:

1. **Multiple structural hypotheses per agent** — a mixture or discrete model index $m \in \{M_1, \ldots, M_K\}$ over wirings, so an agent can be genuinely torn: $q_i(\boldsymbol{\varphi}, m)$, not $q_i(\boldsymbol{\varphi})$.
2. **Divergence as state** — the difference between candidate nets, $D(M_1 \| M_2)$, carried and updated online so an agent can act on *where* two wirings disagree. (Models of other agents' nets are the social special case.)
3. **Communication of hypotheses** — messages that transmit *which structure* an agent entertains (wiring, $\boldsymbol{\Pi}$-pattern), not just means and precisions to be averaged.

Outlook (one line)

Inferred reliability gates (Student-t trust, v4) make divergence *metastable* — but consensus remains the only fixed point. Gates buy time; only represented rivals buy walls.

How much doubt does pluralism need?

Each rung adds model-doubt to the *state*; each unlocks dynamics — and every rung but the last still converges.

Representation	**Doubt in the state**	**Unlocks**	**Still converges because**
Gaussian $(\boldsymbol{\Pi}, \boldsymbol{h})$	none	nothing (null regime)	pooling is a contraction
Student-t gates	doubt about sources	metastable schism	gates buy time, not walls
mixture $q(\boldsymbol{\varphi}, m)$	*precise* doubt about models	rivalry, suspended judgment	Berk: weights concentrate on the KL-closest m[6]
credal set $\{q_m\}$	Knightian doubt	pluralism evidence need not dissolve	it doesn't — that is the point

Precise mixtures are gates by other means: more time, still no walls — permanent pluralism lives on the last rung.

[6] Berk (1966): under misspecification the posterior converges to certainty in the KL-nearest — possibly wrong — model.

Does conviction buy Knightian uncertainty?

Tilt the weights over models: $q(m) \propto p(m \mid o)\, e^{\lambda U(m)}$. For two rivals,

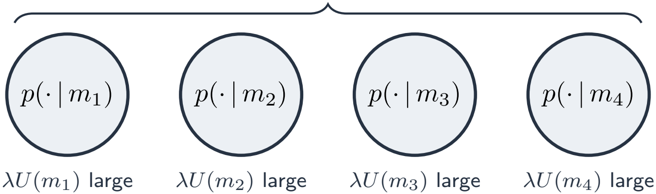
$$\log \frac{q(M_1)}{q(M_2)} = \underbrace{\Lambda_T}_{\text{accumulated log-evidence}} + \underbrace{\lambda\,\Delta U}_{\text{conviction}}$$

- The naive limit fails. $\lambda \to \infty$ in one head $\Rightarrow$ certainty in $\arg\max_m U(m)$: dogmatism, the opposite of held-open ambiguity.
- **Timescale.** Rivals survive while $|\Lambda_T| < \lambda\,\Delta U$, so the merge clock is $T^* \approx \dfrac{\lambda\,\Delta U}{I(o;m)\text{ per obs.}}$ — conviction over the channel capacity for telling models apart: the JSD that the Gaussian erased now sets the clock.
- **Duality (the precise statement).** Exponential tilts are the single-distribution representatives of *set-based* preference families; maxmin Knight is the *pessimistic* member ($e^{-\theta V}$), conviction the *optimistic* one ($e^{+\lambda U}$).[7] Self-evidencing is ambiguity with the sign flipped — Knight is conviction's mirror twin, not its limit.

[7] Hansen & Sargent (2001), multiplier preferences; Maccheroni, Marinacci & Rustichini (2006), variational preferences; Gilboa & Schmeidler (1989), maxmin EU.

Where the credal set actually lives

community state $= \{\, p(\cdot|m_1), \ldots, p(\cdot|m_4)\,\}$: a **literal credal set**, held by no one



↓ precision pooling / federated inference

one precise compromise — the ambiguity is **destroyed**, not resolved

Knightian uncertainty *does* drop out of strong conviction — **heterogeneous** conviction, **across** heads: schools are how a network implements imprecise probability. Expansion (variation), reduction (selection), and conviction (retention) acting on $q(\varphi, m)$ keep the set populated; pooling collapses it.

Summary

1. **No divergence is a theorem, not a bug.** Pooling contracts; Gaussian conflict resolves by confident compromise. The full model *could not* have shown schools or schisms.

2. **The representational slot for disagreement is missing.** $(\boldsymbol{\Pi}, \boldsymbol{h})$ stores one mean and one precision; "torn," "rival," and "who disagrees about what" have nowhere to live.

3. **The concrete next requirement is holding two nets at once.** An agent that can carry *rival Bayes nets and the divergence between them* — within one head or between heads — so that structure-discriminating evidence accumulates instead of being erased at deposit time.

Discussion: what is the minimal representation that supports a represented rival — and should the credal set live in heads, or in the network?